

Piyush Kumar

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ABOUT

Computer Science undergraduate specializing in **Computer Networking** with a strong foundation in **Artificial Intelligence, Machine Learning, Data Structures, and System Design**. Experienced in developing **AI-driven applications, network-based systems and full-stack solutions** through research projects and IoT systems. Skilled in **Java, Python, ML algorithms, and networking technologies**, focused on building scalable and efficient engineering solutions.

EDUCATION

SRM Institute of Science and Technology 2023 – 2027
B.Tech in CSE (Computer Networking) CGPA: 9.83 / 10

T.N.B. College, Bhagalpur 2021
Class XII (BSEB) – PCM 85.4%

D.A.V. Public School, Bhagalpur 2019
Class X (CBSE) – PCMB 89.4%

SOFT SKILLS

- Communication
- Teamwork
- Problem Solving
- Leadership
- Time Management
- Adaptability

EXPERIENCE

King Faisal University Jun 2025 – Present
Research Intern – AI/ML (Solar Energy Prediction) *Online, Saudi Arabia*
– Developed a hybrid regression-based solar energy forecasting system using **5+ years of real climatic data**, improving prediction accuracy by **20%**.
– Implemented ensemble and Bi-LSTM models, reducing forecasting error (RMSE) by **15%** through optimized feature engineering.

SRM Institute of Science and Technology May 2025 – Jul 2025
Summer Research Intern – AI/ML *Kattankulathur, India*
– Enhanced image denoising pipelines for low-dose CT scans, improving signal-to-noise ratio and visual clarity by **20%**.
– Built ML-based pulmonary nodule detection models achieving **85%+ validation accuracy**.

PROJECTS

DSA Visualization Platform [\[Algorithm Visualizer\]](#) Mar 2026 – Present
Next.js, Data Structures & Algorithms, Supabase Auth, REST APIs, Vercel
– Developed an interactive platform to visualize Data Structures & Algorithms with step-by-step animations for sorting, searching, graphs, trees, linked lists, and dynamic programming.

AI Email Reply Generator [\[Email Writer Assistant\]](#) Jan 2026 – Feb 2026
Java, Spring Boot, React.js, Gemini API, Render
– Designed and deployed a full-stack AI email automation system using Spring Boot with secure API integration and production-level error handling.

Smart Home Automation with Jarvis [\[Demo\]](#) Jan 2025 – Jun 2025
Raspberry Pi, ESP8266, Python, IoT, Relay Module
– Implemented a voice-controlled smart home system with real-time monitoring.

SKILLS

- **Programming Languages:** Java, Python, JavaScript
- **Web Development:** React.js, Next.js, HTML5, CSS3, Material UI
- **Backend & APIs:** Spring Boot, RESTful APIs, MongoDB, MySQL, PostgreSQL
- **AI / Machine Learning:** Machine Learning, TensorFlow, Scikit-learn, Large Language Models (LLMs), Prompt Engineering, Retrieval Augmented Generation (RAG)
- **Cloud & DevOps:** AWS, Oracle Cloud Infrastructure (OCI), OCI Generative AI Services, Render, GitHub
- **Networking & Security:** Computer Networking, SD-WAN, Cisco DNA Center, Network Security
- **Tools & Platforms:** Git, Cisco Packet Tracer
- **Hardware / IoT:** Raspberry Pi, ESP8266, Arduino, Sensors

CERTIFICATIONS

- Oracle Certified Professional – Generative AI
- AWS Cloud Practitioner
- NPTEL – Programming in Java

AWARDS & SCHOLARSHIPS

- **Research Nexus Certificate** – 3rd Position
SRM Institute of Science and Technology
- **Certificate of Merit** – 1st Position
Dept. of Physics and Nanotechnology, SRMIST
- **Performance-Based Scholarship** – 2 Times (2024, 2025)
SRM Institute of Science and Technology

LANGUAGES

English (Working Proficiency) Hindi (Native) German (Elementary)